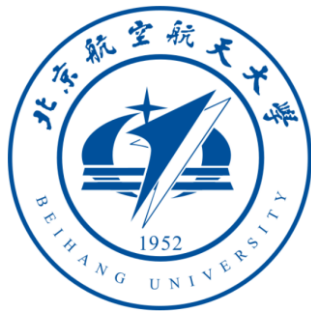




Experimental Study of Ground Reflection and Microphone Installation Effects on Noise Measurements

Jiaju Zou^{1,*}, Lin Du¹, Weiguang Zhang¹, Zhizhen Wang¹

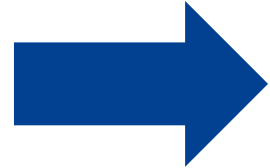
Mingsui Yang², Meng Wang²

¹ Beihang University

² AECC Shenyang Engine Research Institute

32nd AIAA/CEAS Aeroacoustics Conference

Brussels, Belgium, May 26–29, 2026



1

- **Background**

2

- **Experimental Setup**

3

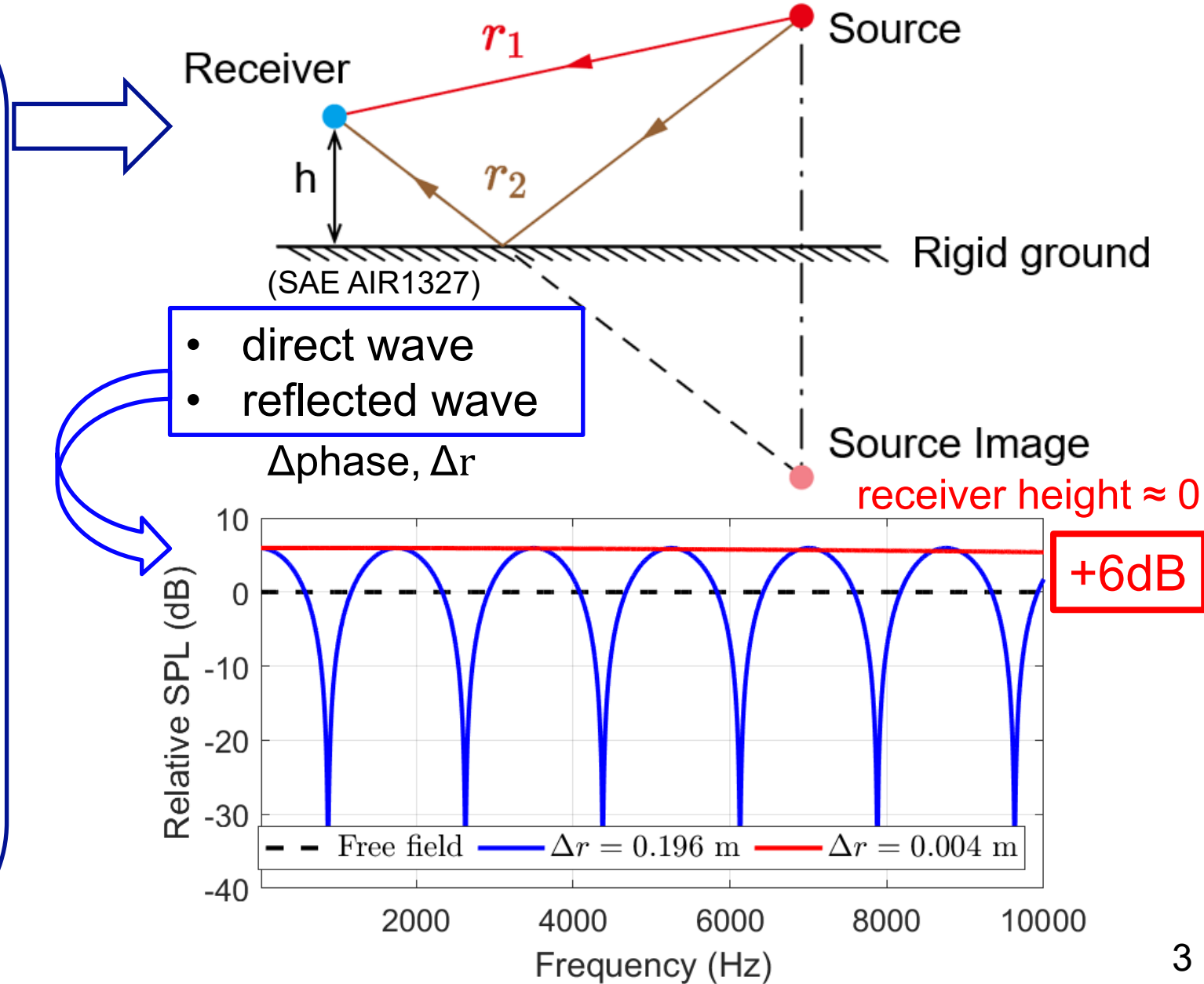
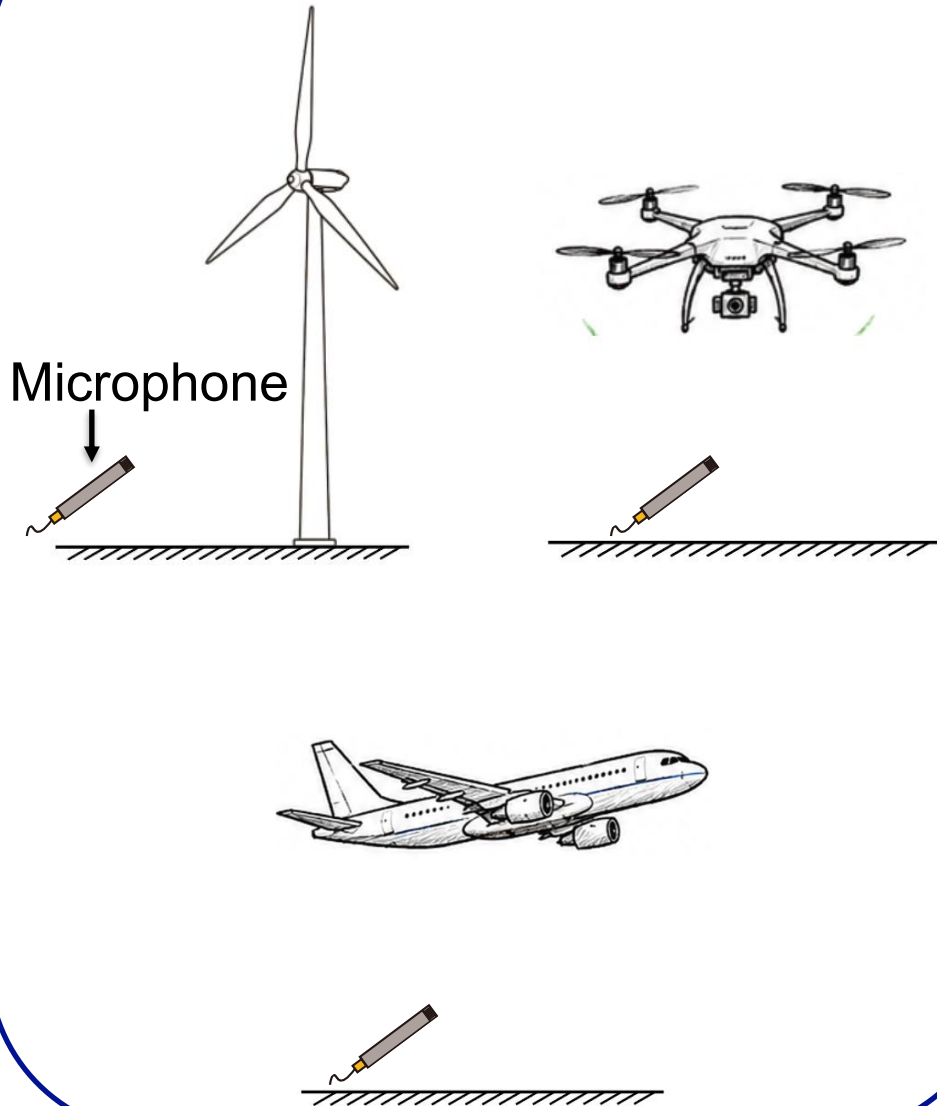
- **Results & Discussion**

4

- **Conclusion**

1. Background

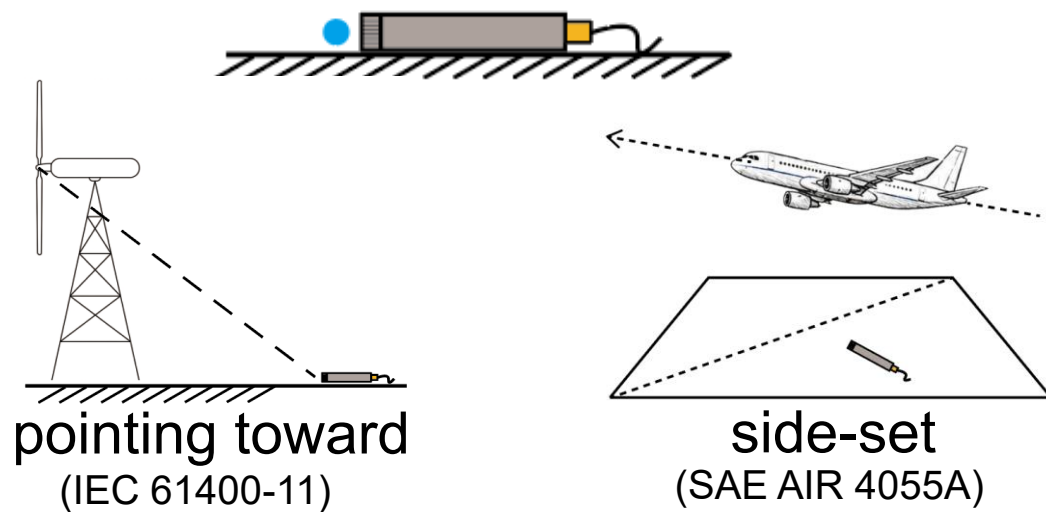
Ground Noise Measurement



1. Background

Two types of ground microphone installation

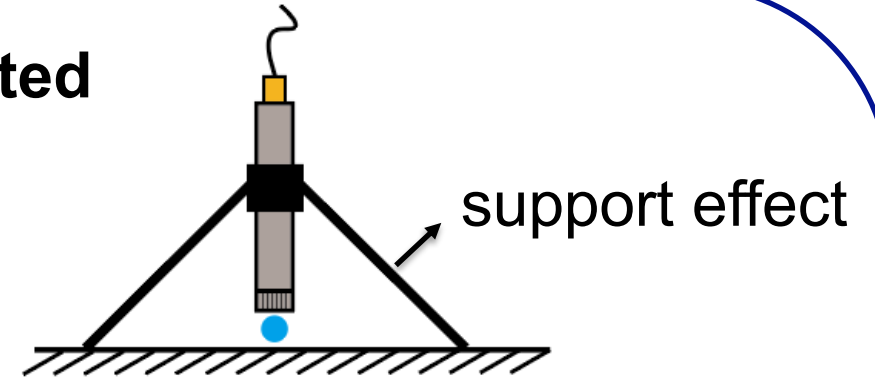
Lying



$$\phi \neq 0^\circ$$

orientation effect

Inverted



support correction

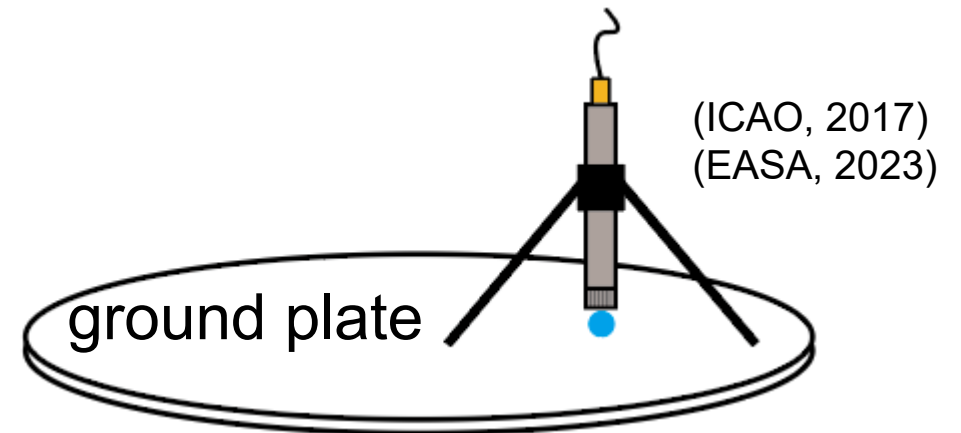


plate response

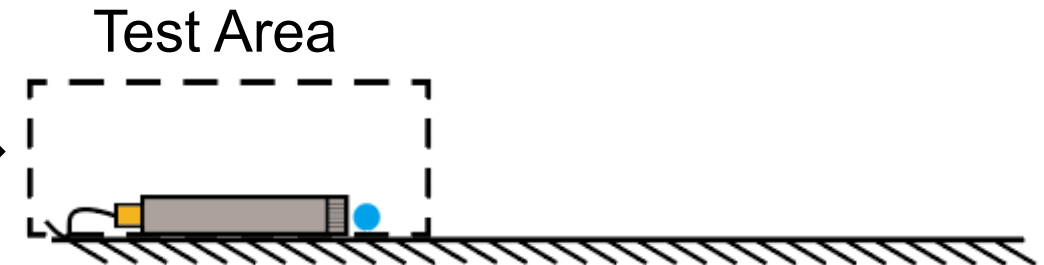
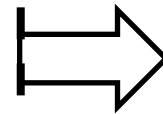
Factors Affecting Ground Noise Measurements

Ground reflection

- ideal correction (+6dB) commonly used
- practical correction remains unclear

Microphone installation

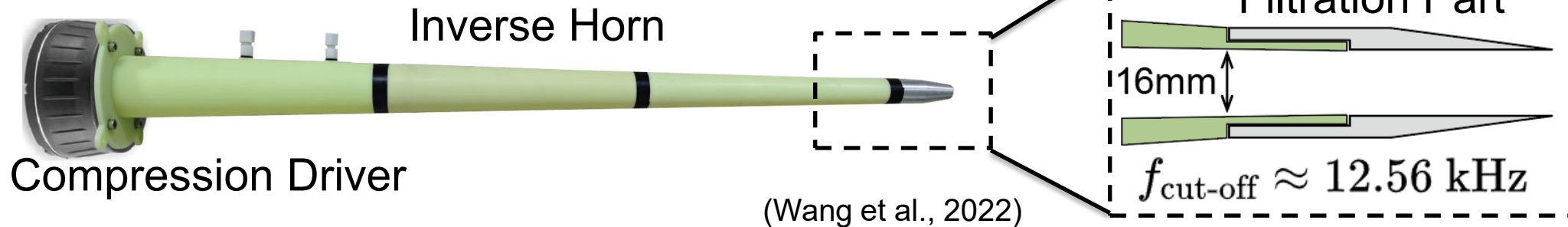
- orientation effect of lying mic
- support correction for inverted mic
- plate response



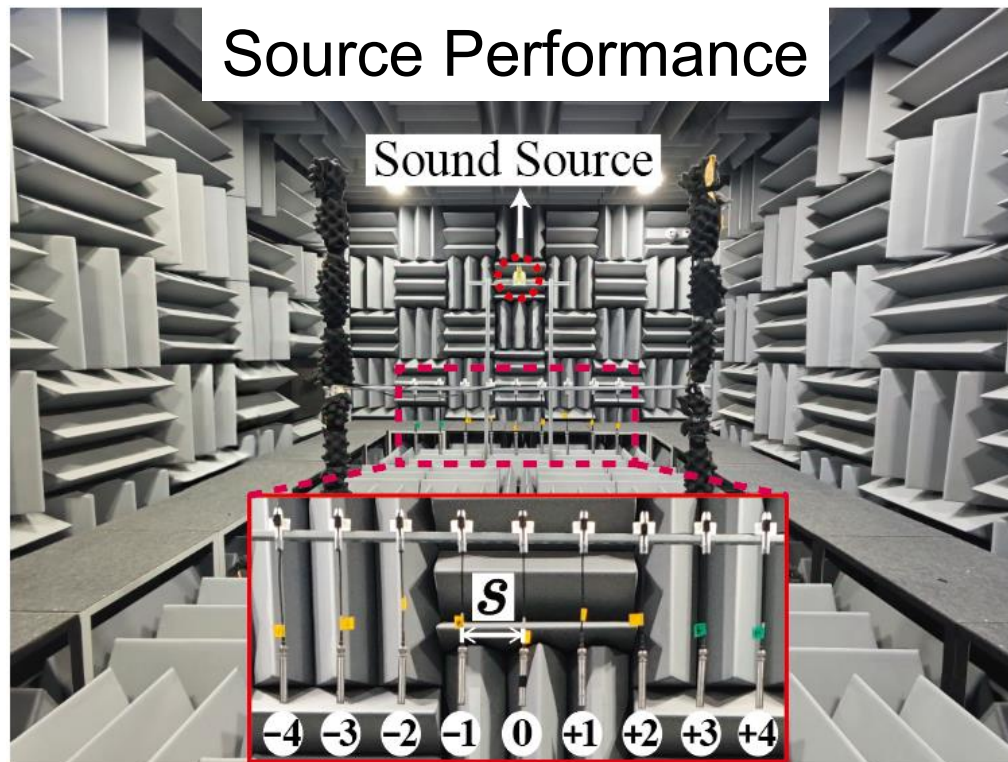
- 
- 1** • **Background**
 - 2** • **Experimental Setup**
 - 3** • **Results & Discussion**
 - 4** • **Conclusion**

2. Experimental Set-up

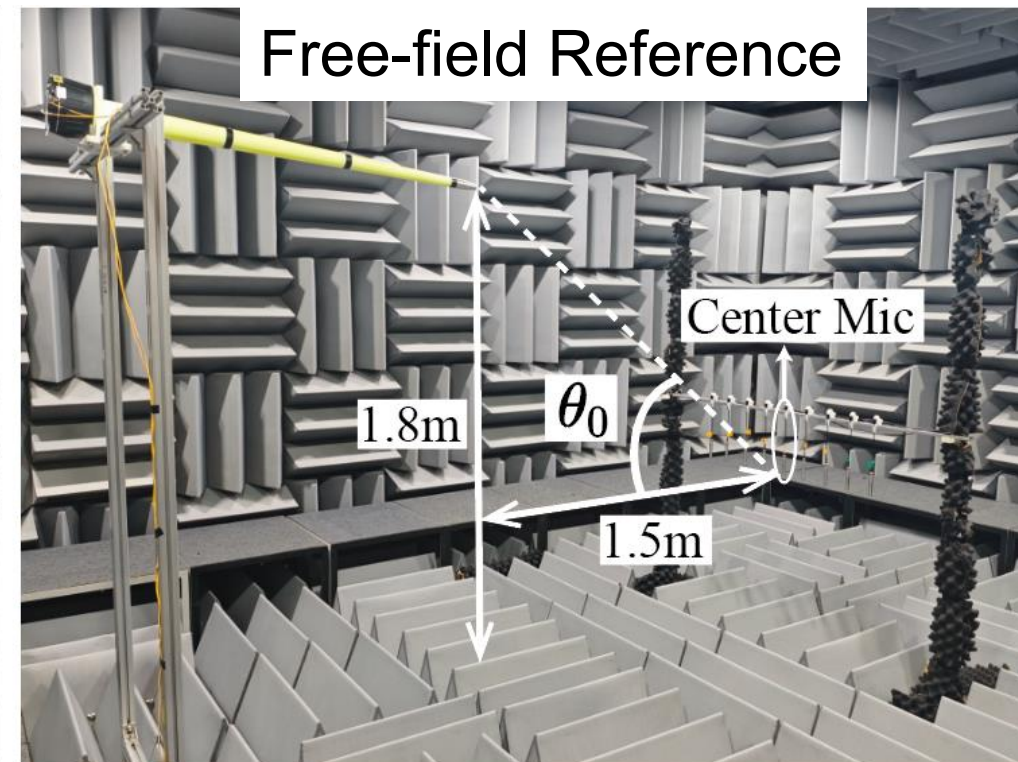
Sound Source



Free-field Set-up



9 suspended microphones



2. Experimental Set-up

Semi-anechoic Set-up

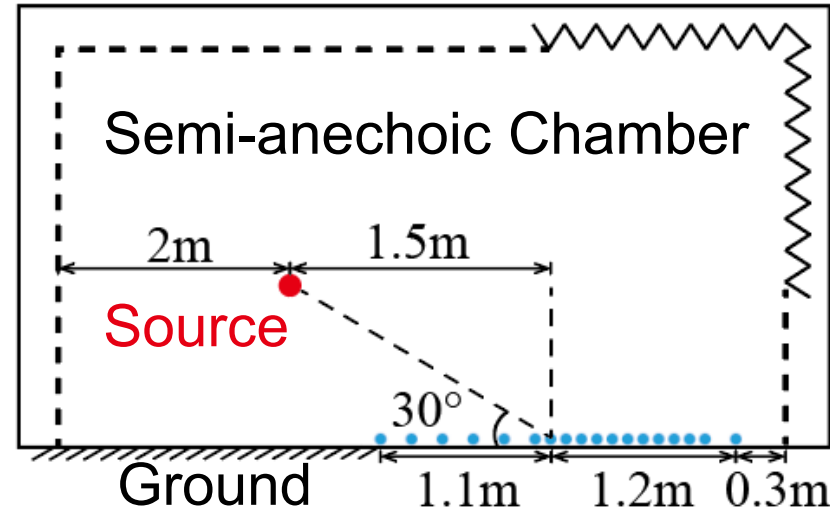
Indoor Test



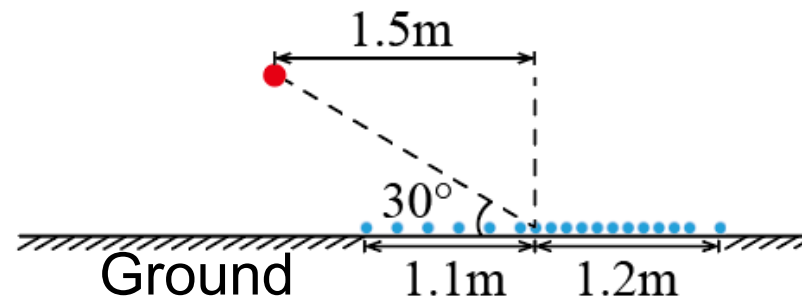
Outdoor Test



Semi-anechoic validation

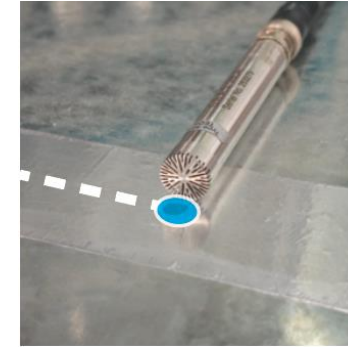


Open Square

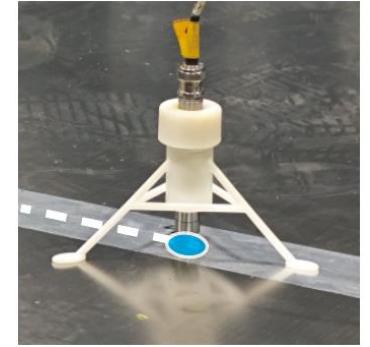


Microphone Installations

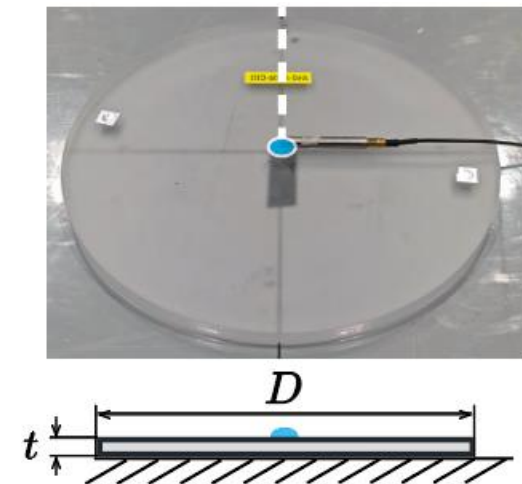
Lying



Inverted



Ground plate



1

- **Background**

2

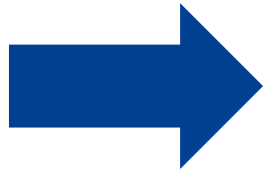
- **Experimental Setup**

3

- **Results & Discussion**

4

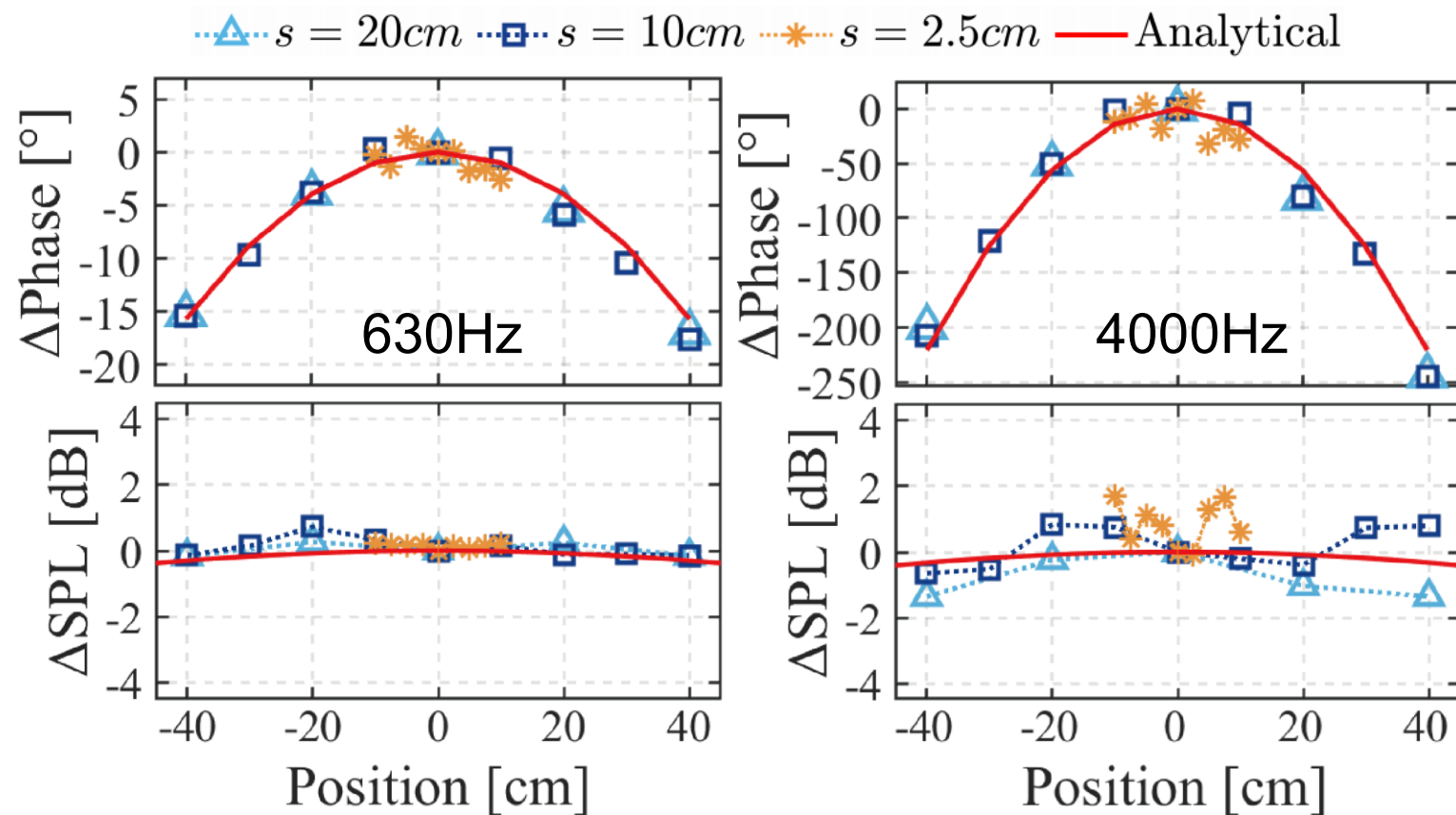
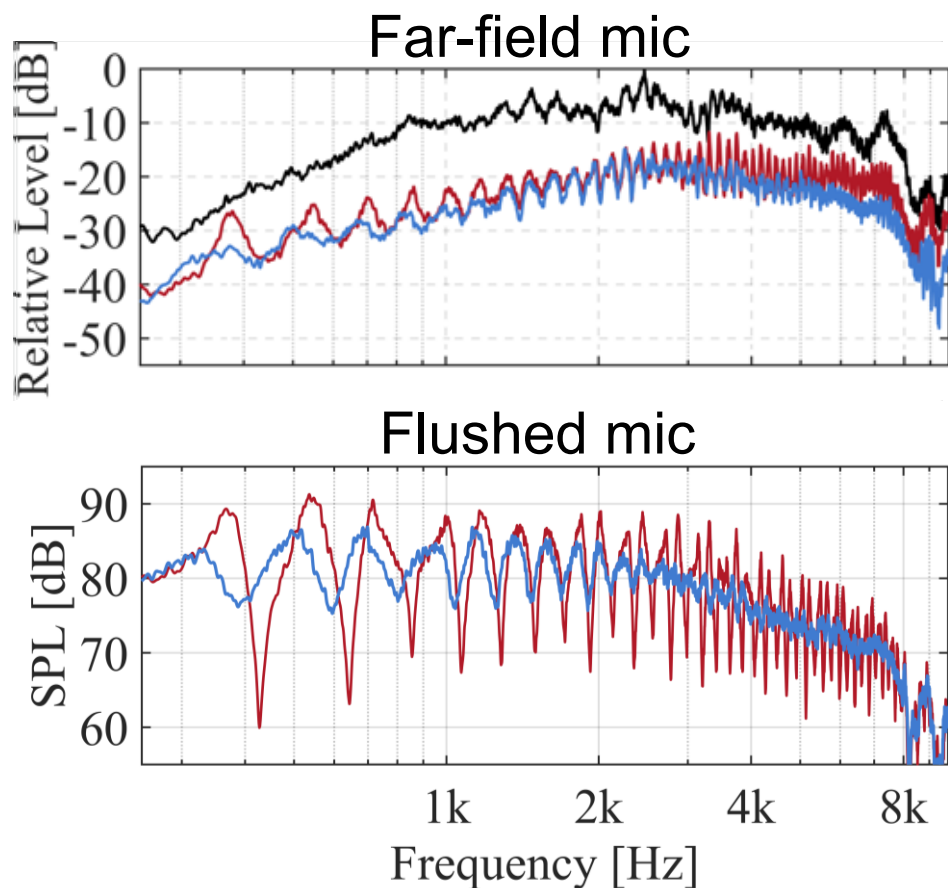
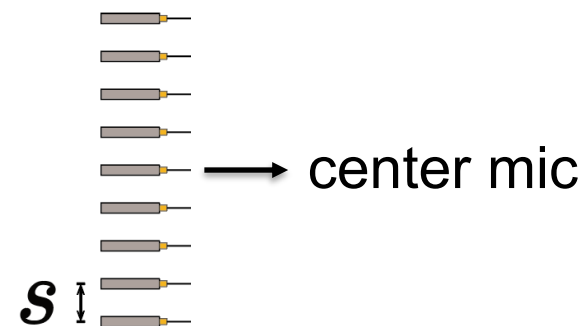
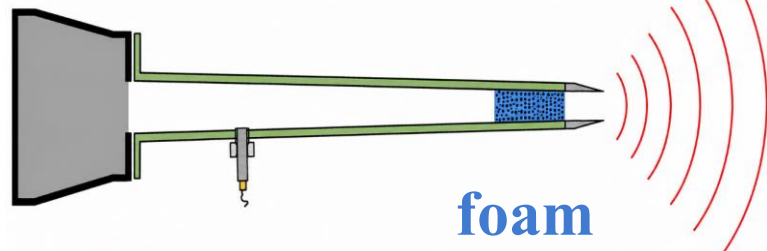
- **Conclusion**



3. Results & Discussion

3.1 Sound Source Performance

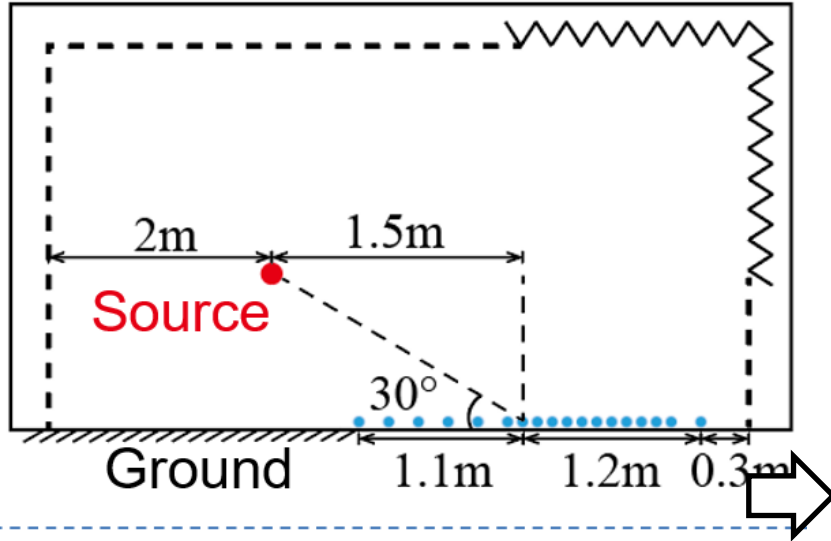
- compression driver
- with inverse horn
- with foam cylinder



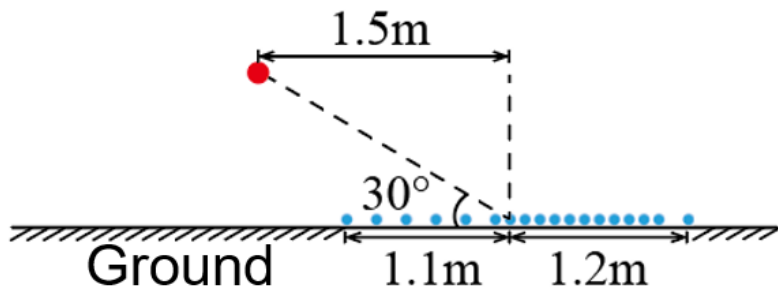
3. Results & Discussion

3.2 Semi-anechoic Condition Validation

Semi-anechoic Chamber

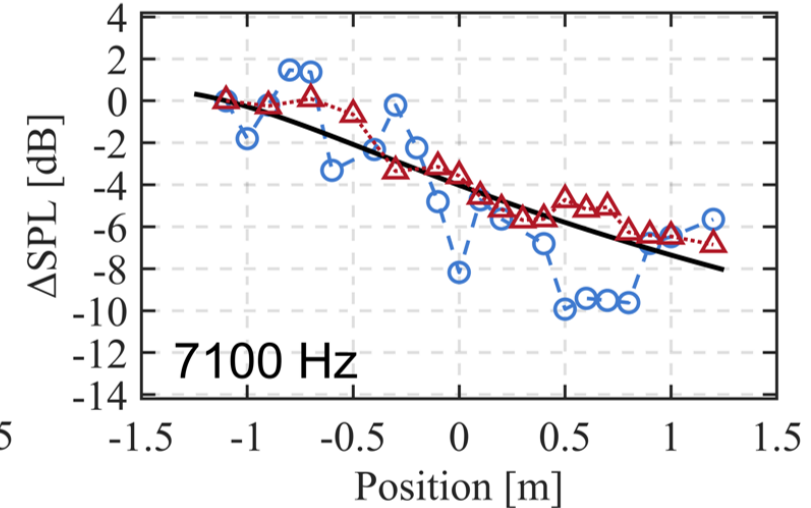
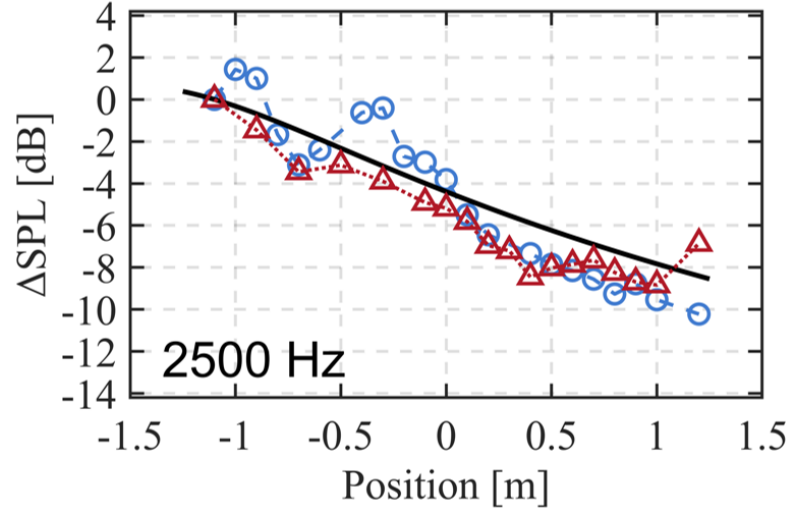
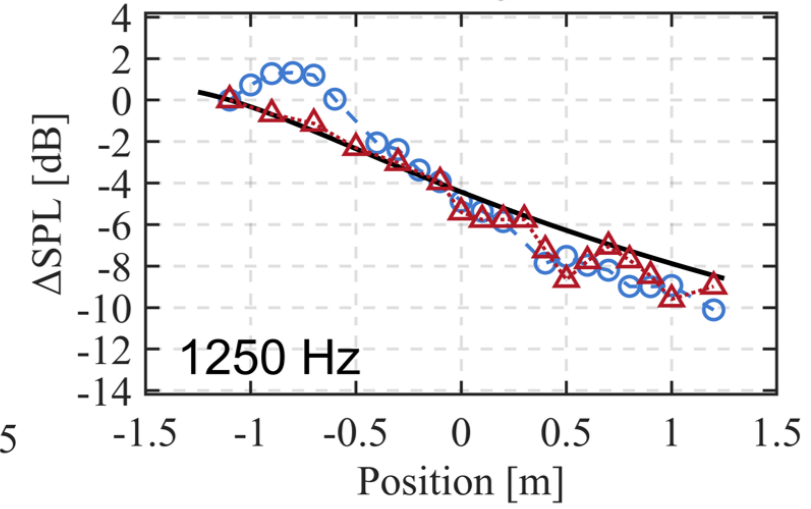
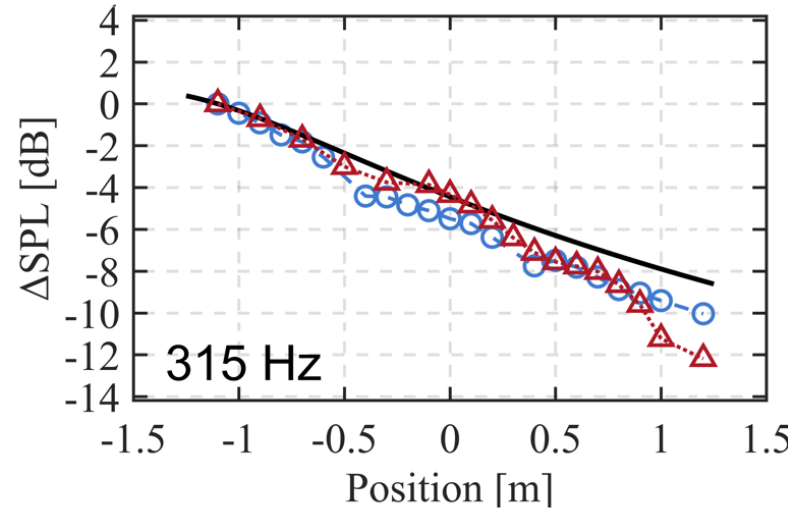


Open Square



- sound source
- measurement point

-△- Indoor test -○- Outdoor test — Analytical

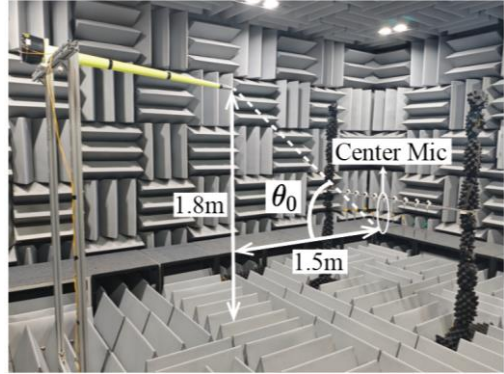


- Uncontrollable outdoor environment
- Valid indoor semi-anechoic condition

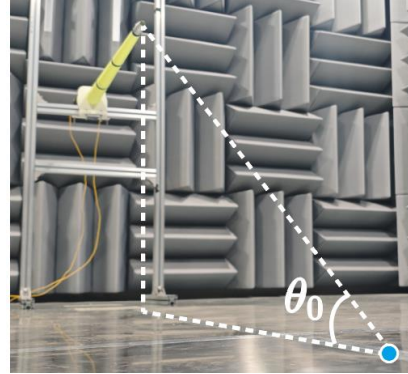
3. Results & Discussion

3.3 Ground Reflection Effect

$$\Delta\text{SPL}_{\text{free-field}} = \text{SPL}_{\text{semi-anechoic}} - \text{SPL}_{\text{free-field}}$$

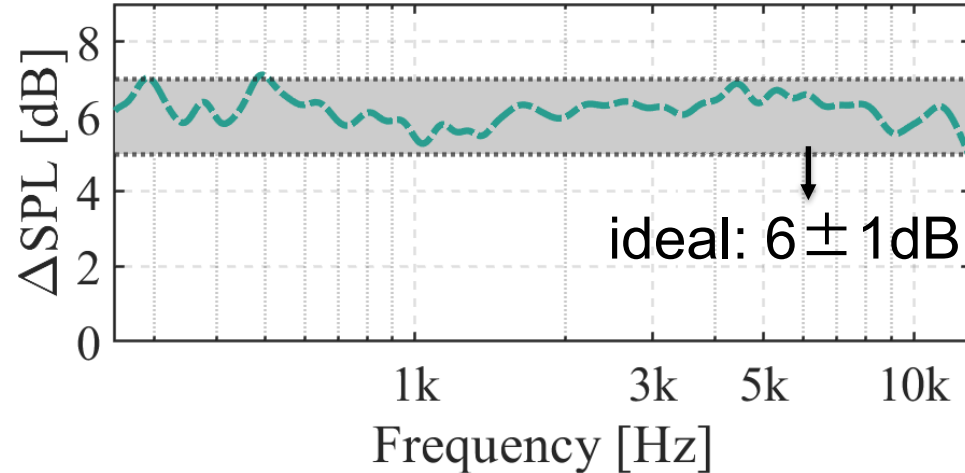


Free-field

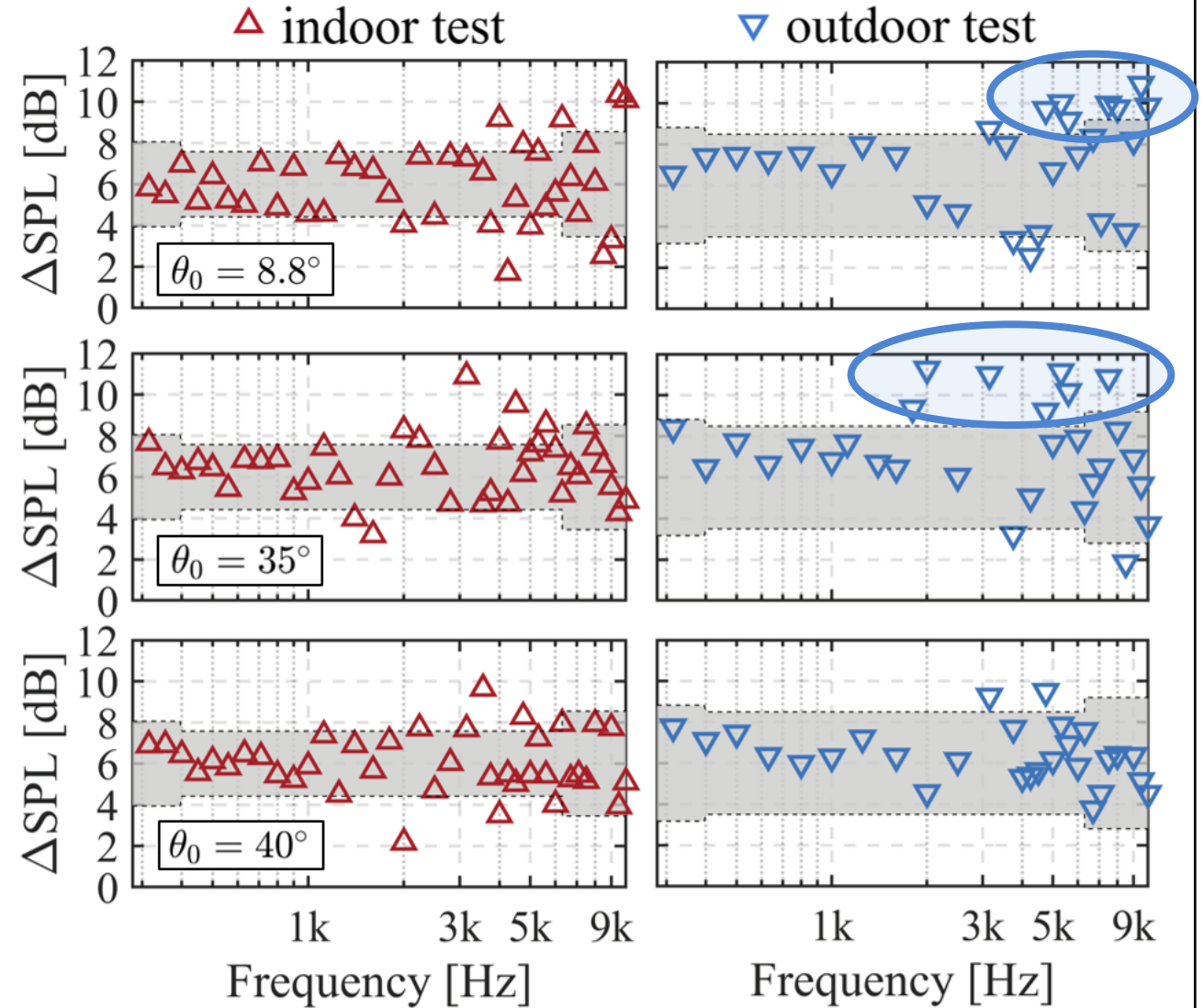


Semi-anechoic

White noise

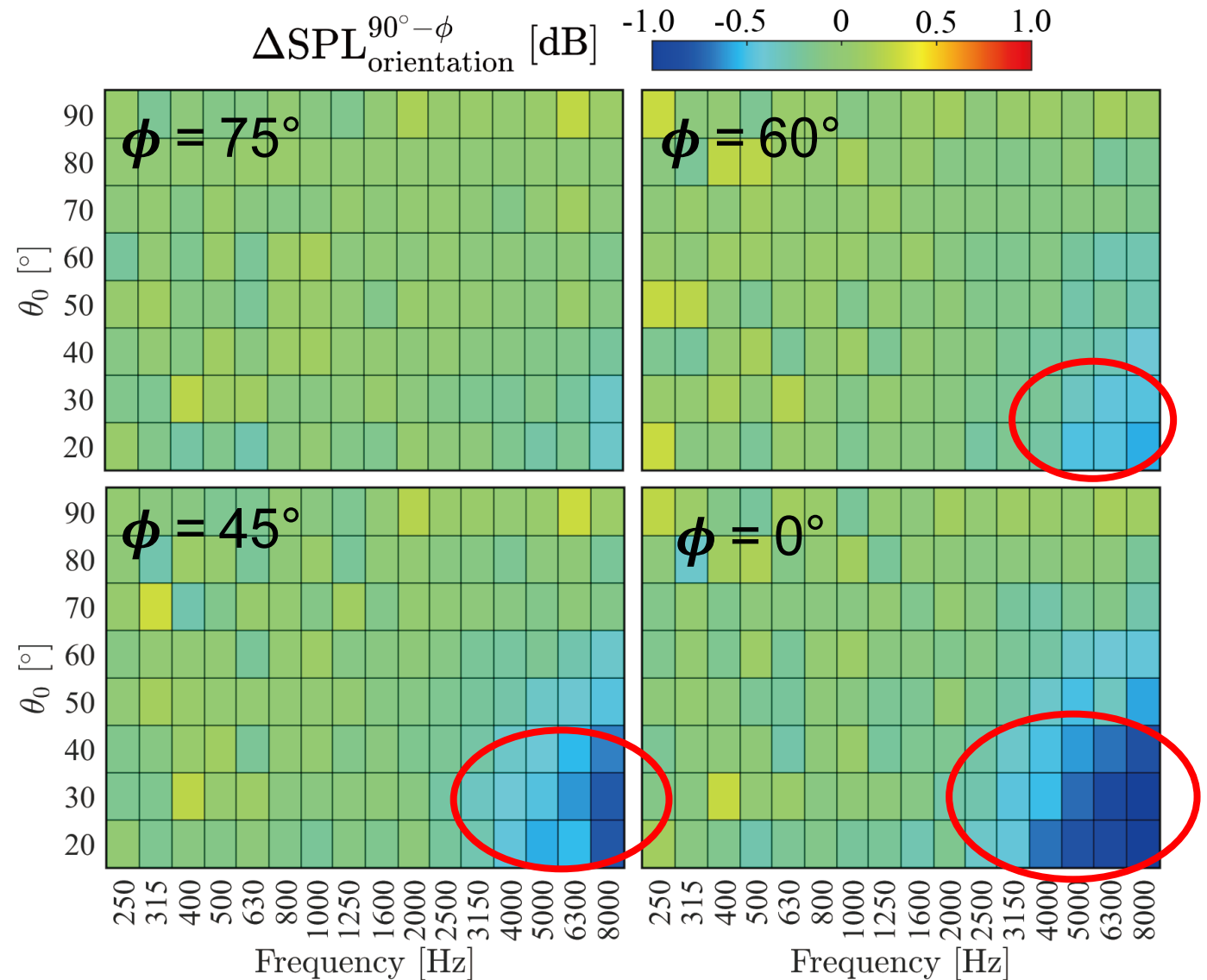
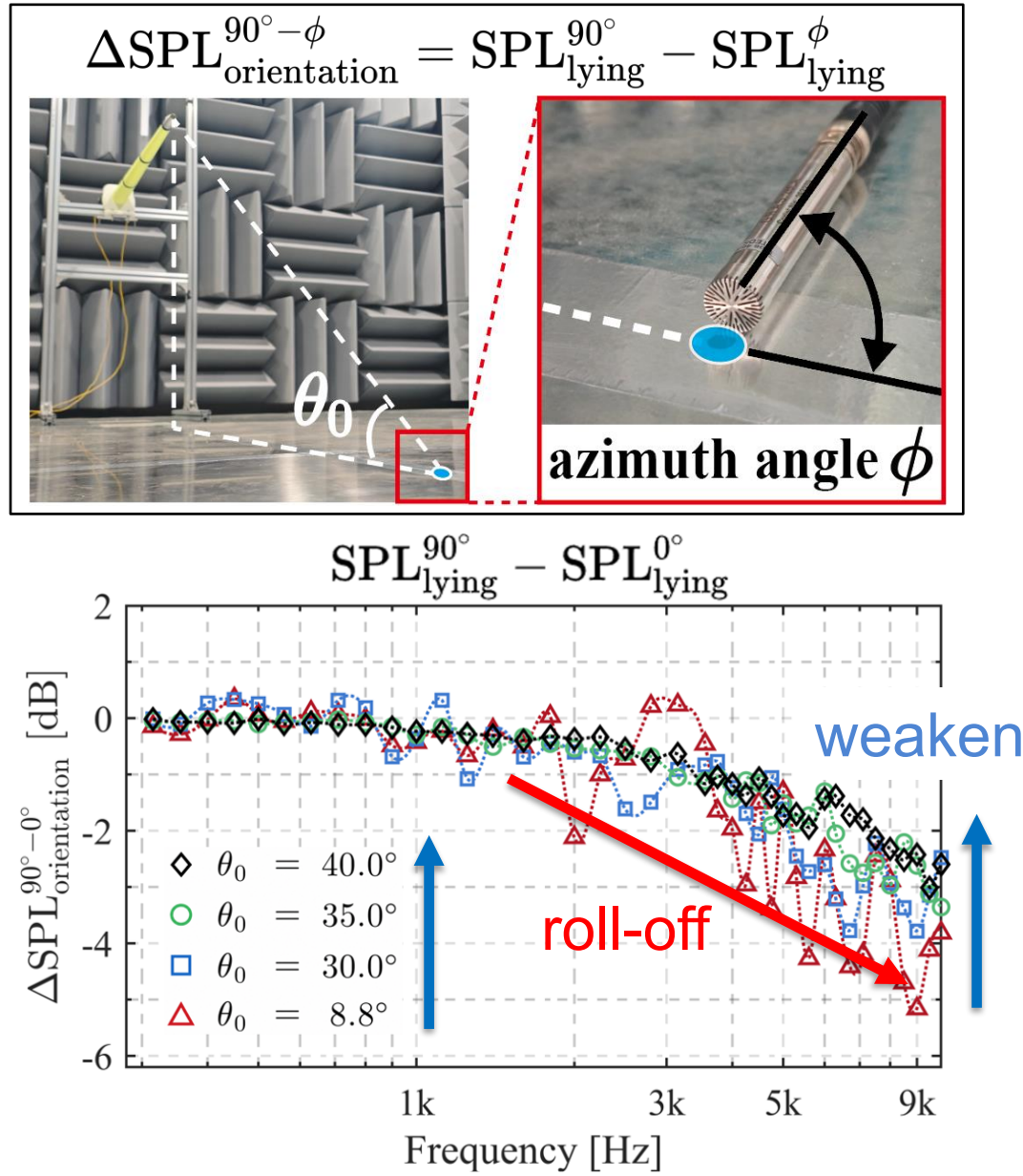


Pure-tone Source



3. Results & Discussion

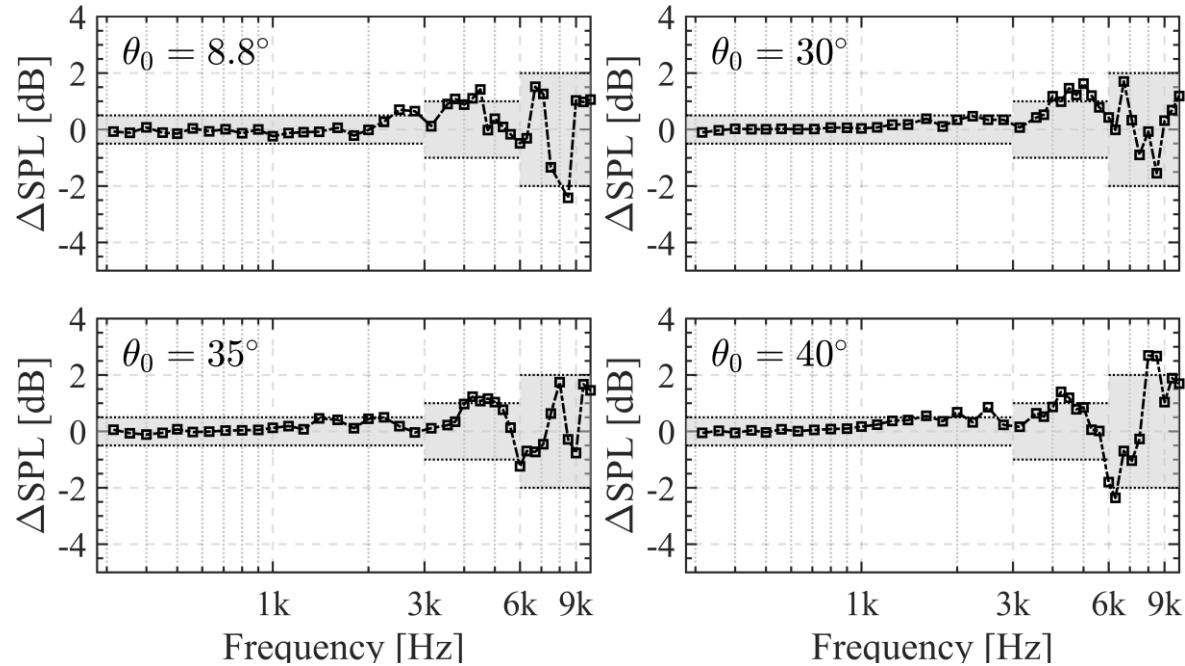
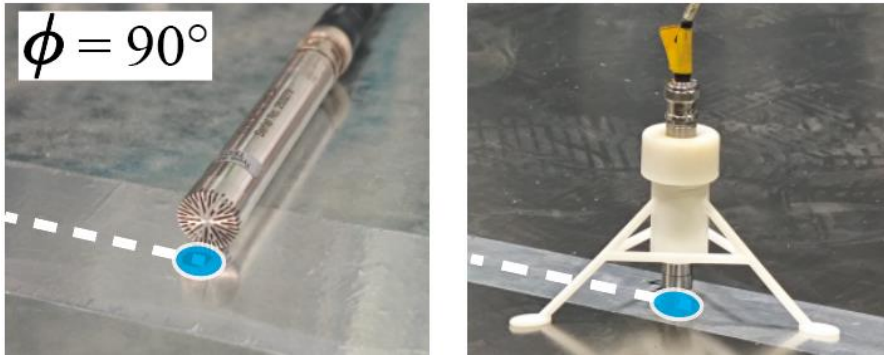
3.4 Microphone Orientation Effect



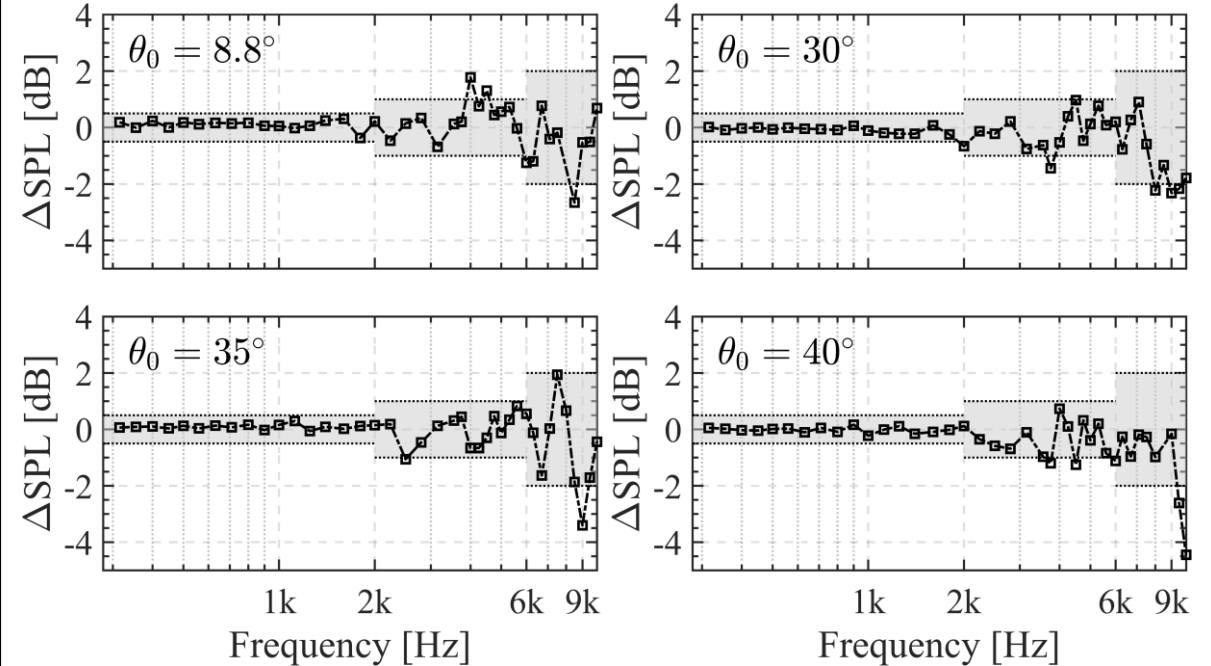
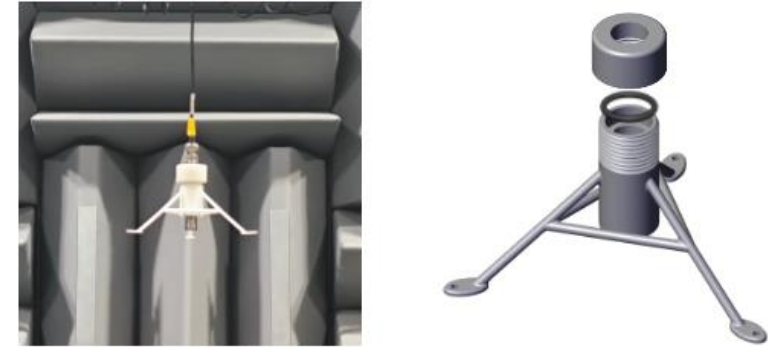
3. Results & Discussion

3.5 Microphone Installation Effect

$$\Delta\text{SPL}_{\text{installation}} = \text{SPL}_{\text{inverted}} - \text{SPL}_{\text{side-set}}$$

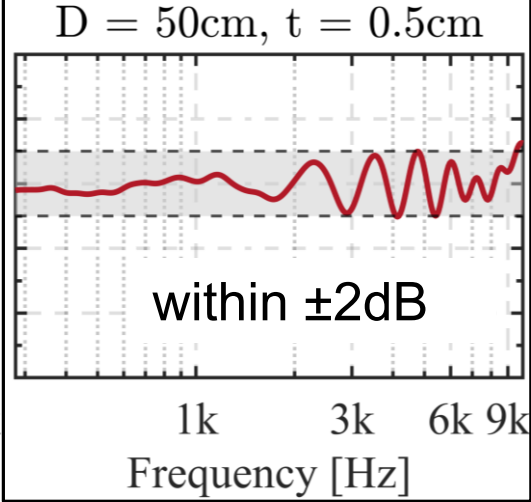
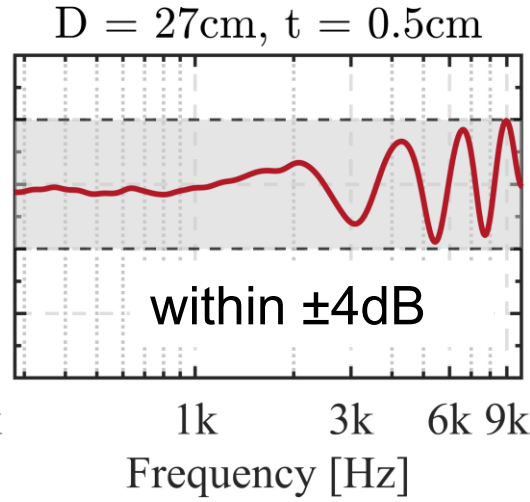
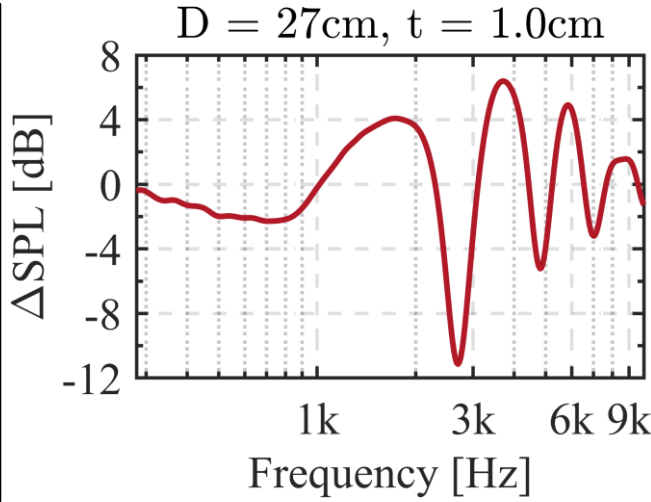
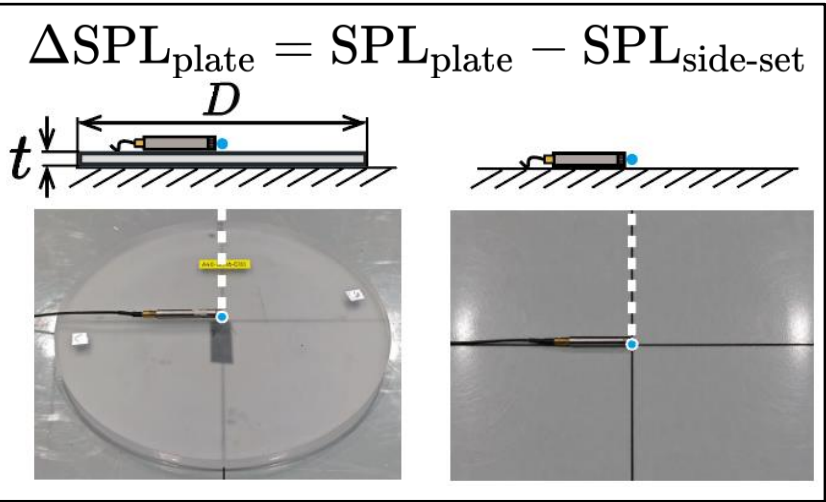


$$\Delta\text{SPL}_{\text{support}} = \text{SPL}_{\text{w/ sup.}} - \text{SPL}_{\text{w/o sup.}}$$

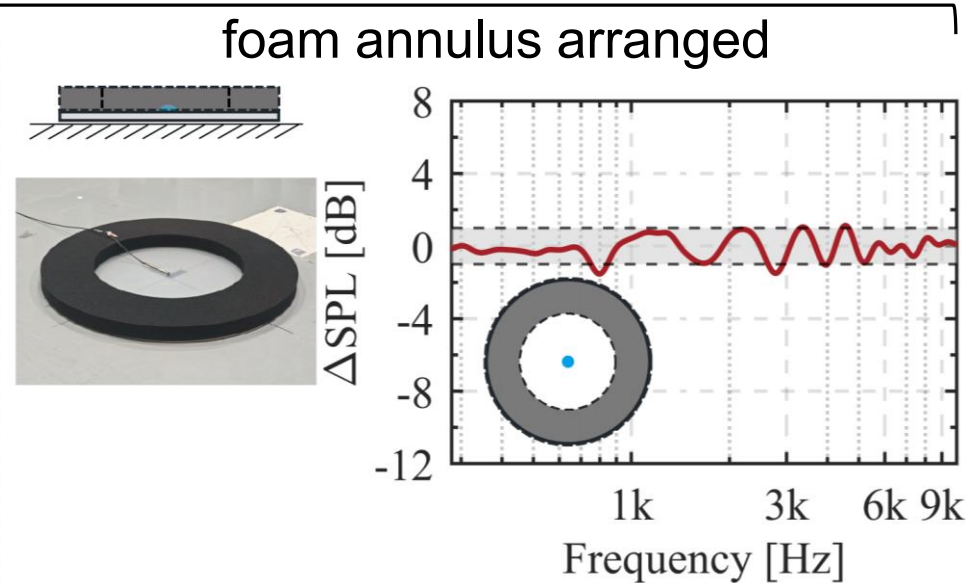
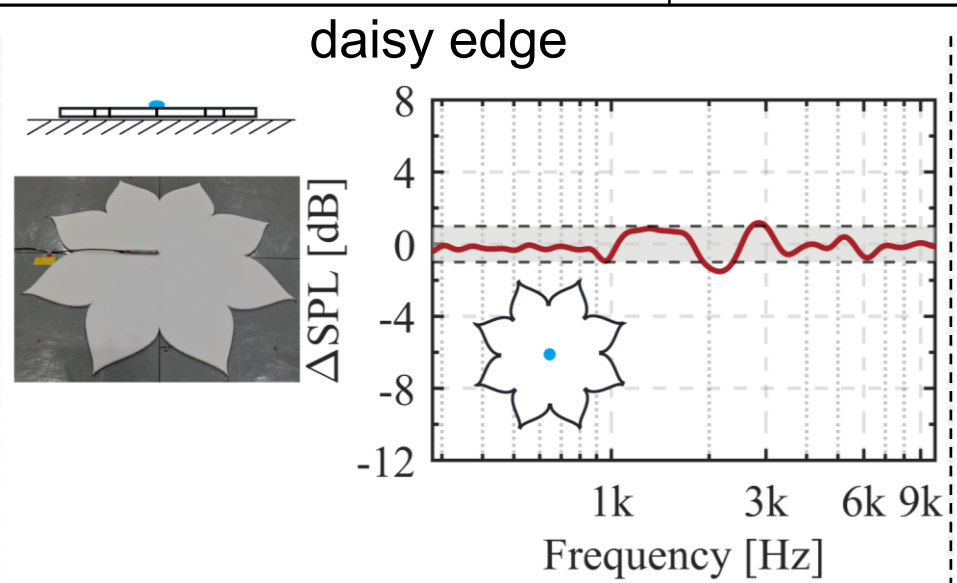
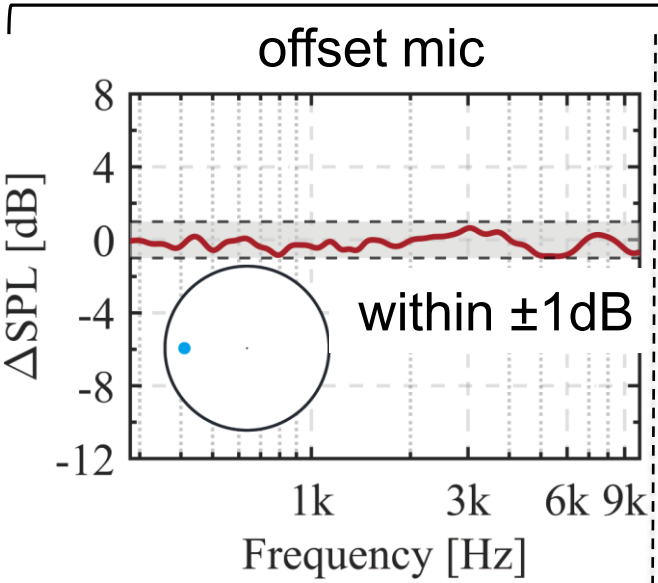


3. Results & Discussion

3.6 Ground Plate Response



mitigate edge effects



A potential approach

1

- **Background**

2

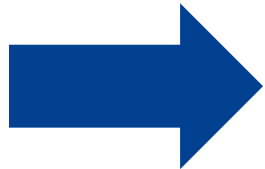
- **Experimental Setup**

3

- **Results & Discussion**

4

- **Conclusion**



4. Conclusion & Outlook



Ground Reflection

Indoor measurements

white noise: stay within $6 \pm 1\text{dB}$

pure tones: follow $6 \pm 1\text{dB}$ trend

(local deviations of 2–3 dB)

Outdoor measurements

uncontrolled environment

irregular deviations (max $\sim 5\text{dB}$)

The ideal $+6\text{dB}$ value is not universally applicable and requires practical correction.

Microphone Installation

Lying microphone:

diaphragm exposure $\uparrow$, measured SPL $\uparrow$

Inverted microphone:

identical to the side-set mic results ($< 3\text{ kHz}$)

support affect noise measurement ($> 2\text{ kHz}$)
(max $\sim 2\text{dB}$)

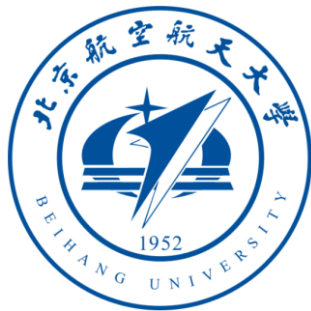
Ground plate:

thickness-dependent edge effect

potential mitigation method: foam annulus

Require minimal microphone support

Mechanism of the foam annulus



Q&A

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